7. Project Documentation Phase

Script-Controlled ACL: Restrict Record Access Based on Field Value

Project Summary

This lab demonstrates how to implement a **script-controlled Access Control List** that restricts users from viewing records unless they meet specific conditions. Only users of the **EEE branch** can view records whose **Branch** field is EEE, while administrators retain full access. Estimated duration: 3 hours.

Configuration Summary

Item	Value
Test user	EEE User (eeeuser@gmail.com)
Roles	bb1, bb2, bb3, bb4
Table	Institution Details (u_institution_details)
Fields	Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description
Branch choices	ECE, EEE, CSE

ACL Summary

Operation	Role	Condition	Behaviour
Read	bb1	Branch is EEE + script	View only EEE records; admin sees all
Create	bb2	None	New button available
Write	bb3	None	Records can be edited
Delete	bb4	None	Records can be deleted

ACL Script

```
(function () {
    // Allow admin users full access
    if (gs.hasRole('admin')) {
        return true;
    }
    // Allow only EEE branch users to see EEE records
    if (gs.hasRole('bb1')) {
        return true;
    }
    // Deny access for all others
    return false;
})();
```

Verification Summary

- bb1 holders view only EEE branch records; users without the role see no records; admin sees everything.
- bb1 + bb2 adds the New button; bb1 + bb2 + bb3 adds editing; bb1 to bb4 adds deletion.

Learning Outcome

Learners gain a comprehensive understanding of how script-controlled READ, WRITE, CREATE and DELETE ACLs enforce record-level security, how user roles and record field values are evaluated together to make access decisions, and how ACLs protect sensitive data from unauthorized viewing, modification, creation and deletion across forms, lists and Service Portal interfaces.